

DSA STUDY BLUEPRINT SERIES

20. CPP STL Complete Tutorial

Containers, Iterators, and Algorithms in Standard Template Library

Section 03 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **STL:** Provides templates for standard data structures (vector, list, stack, queue, map, set).
- **Algorithms:** Built-in optimized utilities like `sort()`, `reverse()`, and `binary_search()`.

Memory & Execution Blueprint

Containers
(Data)

↔ Iterators
(Pointers) ↔

Algorithms
(Logic)

C++ Implementation Reference

Standard code layout and syntax

```
#include  
#include  
#include  
  
vector v = {3, 1, 2};  
sort(v.begin(), v.end()); // sorts vector  
unordered_map mp;  
mp[1] = "one";
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
STL Sort	$O(N \log N)$ (IntroSort)
Map insertion	$O(1)$ average / $O(\log N)$ ordered

Key Practices & Gotchas

- **Important:** Iterators act as glue connecting containers with algorithms.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace how standard templates process vector insertions and compiler tasks:

Phase / Action	Execution Step	Underlying Resource Allocation	Expected Output
Vector insert	<code>v.push_back(10);</code>	Allocates new contiguous heap memory block	Element added at back
GCC Compilation	<code>g++ -std=c++17</code>	Parses syntax AST and creates system binary	Native executable binary file

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Container Mechanics:** Vectors double heap size automatically; lists use node nodes; maps build red-black search trees.
- **Build Warnings:** Use `-Wall -Wextra` flags to enforce strict pointer checks and variables usage during compiler tasks.
- **Related LeetCode Problems:** #1 Two Sum (using `unordered_map`), #20 Valid Parentheses (using stack).